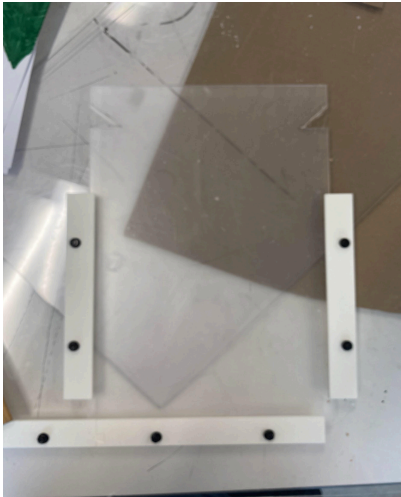
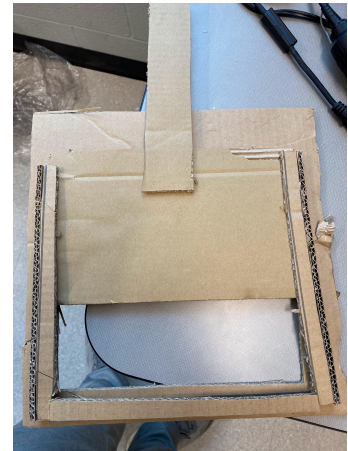
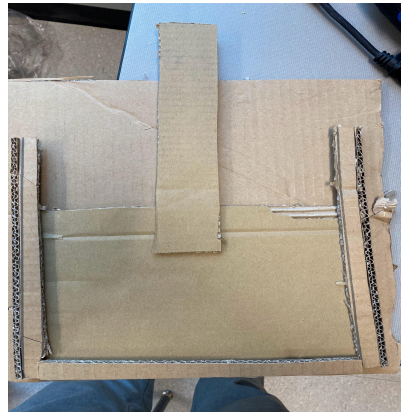


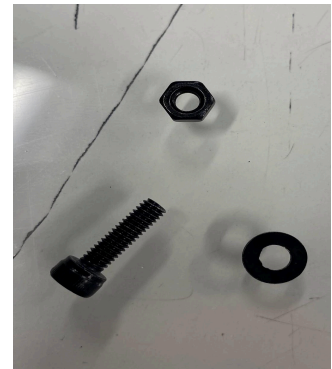
Construction of a testable prototype

We will work with the other groups on the project to construct a testable prototype. The initial testable prototype was made of cardboard to test sliding systems and handle designs. These photos show the parts and construction of the initial prototype. The first was made of cardboard. This was made to get an idea of a shape and work with different designs and ideas from this.



We then used a sheet of clear acrylic as a base for the next prototype. We designed and 3D-printed tracks for the door to slide on. Once printed out, we drilled holes into the acrylic to mount the tracks and bolted them on the acrylic with **small bolts** (As pictured to the right).

We then used a smaller piece of acrylic to make the door for the prototype (Picture on the left). We began to think of different ways to have a latching mechanism that will not interfere with the X-ray. We concluded that a ratcheting system would be most beneficial for action. After



talking with experts and among the groups, we also decided to create a complaint mechanism so that the latch system wouldn't affect the ability for the zookeepers to x-ray through the sides. We furthered that we would attach the tracks to the box with nylon bolts. As we continued to consult with experts, we discussed the nylon's strength and decided it would be stronger if we used more or larger nylon bolts. This allowed for the x-rays to be minimally affected, retaining strength for the doors and sliding system.



When constructing the assembly for the lateral (side) doors to permit

